

Advanced Database

CSIT — 8th Semester — 2079 — Answer Sheet
Subject Code: CSC461

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Section A

1. Design an EER model for library management system having generalization and specialization hierarchies. The EER model should have disjoint and overlapping constraints with at least one of the entities having total participation. Use your own assumption for another concepts. Now convert so designed EER diagram into its equivalent relational model.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. What are different concepts and features of object-oriented database? What is object relational model?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. Define distributed database. What are the benefits of using distributed database over centralized database? Explain availability, reliability and scalability features of distributed databases.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Section B

Attempt any Eight questions[8×5=40]

4. Why important to store data in database? What is multilevel index?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. What is aggregation? Explain with example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. Define ODMG object model. What is object Query language (OQL)?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. What is query processing? Differentiate query processing with query optimization.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. Why query optimization is necessary? Illustrate on the choice of query execution plans.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. Why fragmentation is carried out? Differentiate between horizontal and vertical fragmentation.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. Why NOSQL system is essential? What are different characteristics of this system?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. What is spatial database? Explain the concept of trigger with an example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. Write short notes on a. CAP Theorem b. Deductive database

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.